

UNIT - 1

Definition

- Statistics is a Scientific method,
- Statistics is the science of collecting, organizing, analysing and interpreting data in order to make a decision.

Characteristics

- All the data have to be arranged into homogeneous groups by the process of classification.
- The classification are based on unity in diversity.
- It may be either real (or) imaginary.
- It should be flexible to accept the adjustments.
- It may depend on either similarities (or) dissimilarities.

D/W b/w Primary & secondary data.

	Primary data	Secondary data
1.	It is original since it is collected by the investigator itself.	It is not original because it is collected by some agency and is used by another agency.
2.	A large amount of time, money, labour, planning are needed.	No need of spending much labour, money and time.
3.	It is suitable to the main & objectives of its collection.	The objective of secondary data may vary in later usage.
4.	No special care is needed while using it.	Before using this data, careful observation of the source, mode and objective of data is needed.

Discuss the essential Characteristics of a good average

14:20 ✓

A good average should have the following essential characteristics:

1. **Representativeness:** It should be representative of the data set, capturing the central tendency or typical value.
2. **Uniqueness:** There should be only one value that satisfies the definition of the average.
3. **Stability:** It should be stable and not unduly influenced by extreme values or outliers.
4. **Usefulness:** It should be useful for further analysis or decision-making.
5. **Ease of computation:** It should be easy to calculate and understand.

Some common types of averages include:

- Arithmetic Mean (AM)
- Geometric Mean (GM)
- Harmonic Mean (HM)
- Median
- Mode

Each type of average has its own strengths and weaknesses, and the choice of which one to use depends on the specific context and data characteristics.



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Dispersion: Dispersion refers to the spread or variability of data points within a dataset. It measures how much individual data points deviate from the central tendency (e.g., mean, median, mode).

Objectives of studying dispersion:

1. **Understand data variability:** To understand the extent of variation in the data.
2. **Assess data reliability:** To assess the reliability of the data and the central tendency measures.
3. **Compare datasets:** To compare the variability of different datasets.
4. **Identify outliers:** To identify outliers or unusual observations.
5. **Inform decision-making:** To inform decision-making and predict future outcomes.
6. **Improve process control:** To improve process control and reduce variability in industrial or business processes.

Common measures of dispersion include:

- Range
- Variance
- Standard Deviation
- Interquartile Range (IQR)

Skewness:

Skewness refers to the asymmetry or lack of symmetry in a probability distribution. A skewed distribution is one where the majority of the data points are concentrated on one side of the mean, with a longer tail on the other side.

Types of skewness:

1. **Positive skewness (right-skewed):** The tail is longer on the right side, indicating a few extreme values above the mean.
2. **Negative skewness (left-skewed):** The tail is longer on the left side, indicating a few extreme values below the mean.
3. **Symmetric distribution:** The distribution is symmetrical around the mean, with no skewness.

Importance in statistical analyses:

1. **Assumption checking:** Many statistical tests assume normality (symmetry) of the data. Skewness can indicate non-normality, which may require data transformation or alternative tests.
2. **Data interpretation:** Skewness can affect the interpretation of central tendency measures (mean, median, mode).

Types of Statistics:

There are two main types of statistics: Descriptive Statistics and Inferential Statistics.

1. Descriptive Statistics:

Descriptive statistics summarize and describe the features of a dataset. They provide an overview of the data, including measures of central tendency, dispersion, and distribution shape.

- **Measures of central tendency:** mean, median, mode
- **Measures of dispersion:** range, variance, standard deviation
- **Distribution shape:** skewness, kurtosis

Examples:

- Calculating the average height of a population
- Describing the distribution of scores in a class
- Summarizing customer satisfaction ratings

2. Inferential Statistics:

Inferential statistics use sample data to make inferences or predictions about a larger population. They help us draw conclusions beyond the data we have.



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- **Hypothesis testing:** testing assumptions about a population
- **Confidence intervals:** estimating population parameters
- **Regression analysis:** modeling relationships between variables

Examples:

- Estimating the average income of a city based on a sample survey
- Testing whether a new drug is effective in treating a disease
- Analyzing the relationship between price and demand for a product

Uses of Statistics:

1. **Data analysis:** statistics help us understand and summarize data
2. **Decision-making:** statistics inform business, policy, and personal decisions
3. **Research:** statistics are used in scientific research to test hypotheses and develop theories
4. **Quality control:** statistics are used to monitor and improve process quality

These are just a few examples of the many uses of statistics.

Illustrate the different method of presenting statical data using diagram and graphs

14:57 ✓

Statistical data can be presented using various diagrams and graphs to effectively communicate insights and trends. Here are some common methods:

1. **Bar Chart:** Compares categorical data using rectangular bars of varying heights or lengths.
2. **Pie Chart:** Shows proportion of categories in a dataset, with slices representing percentages.
3. **Histogram:** Displays continuous data distribution using frequency or density plots.
4. **Line Graph:** Illustrates trends over time of continuous variables.
5. **Scatter Plot:** Visualizes relationships between two variables.
6. **Box Plot:** Summarizes data distribution using median, quartiles, and outliers.

Let's take an example dataset:

Month	Sales
Jan	100
Feb	120
Mar	150

Apr	180
May	200

Bar Chart:

<> code
[View code](#)

```
Jan | #####
Feb | #####
Mar | #####
Apr | #####
May | #####
```

Line Graph:

<> code
[View code](#)

```
200 | *
180 | *
150 | *
120 | *
100 | *
    Jan Feb Mar Apr May
```

Pie Chart:

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```
+-----+
| Jan (10%) |
+-----+
```

3. Compare arithmetic mean, geometric mean, and harmonic mean in terms of applicability

14:59 ✓✓

Here's a comparison of Arithmetic Mean (AM), Geometric Mean (GM), and Harmonic Mean (HM) in terms of applicability:

Mean	Applicability	Example
Arithmetic Mean (AM)	Suitable for additive data, normally distributed data	Average salary, average height
Geometric Mean (GM)	Suitable for multiplicative data, skewed distributions, growth rates	Investment returns, population growth
Harmonic Mean (HM)	Suitable for rates, ratios, reciprocals	Average speed, average rate of return

Key differences:

- **AM:** sensitive to extreme values, suitable

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Key differences:

- **AM:** sensitive to extreme values, suitable for linear data
- **GM:** robust to extreme values, suitable for exponential data
- **HM:** suitable for rates, gives more weight to smaller values

Choose the right mean based on your data type and analysis goal!

4. Critically evaluate the role of standard deviations in interpreting variability

15:00 ✓✓

Role of Standard Deviation (SD) in Interpreting Variability:

1. **Measures dispersion:** SD quantifies the spread of data from the mean, indicating variability.
2. **Sensitive to outliers:** SD is affected by extreme values, which can skew interpretation.
3. **Assumes normality:** SD is most effective for normally distributed data; non-normal data may require alternative measures.
4. **Interpretation challenges:** SD is scale-dependent, making comparisons across datasets tricky.
5. **68-95-99.7 rule:** In normal distributions, about 68% of data falls within 1 SD, 95% within 2 SD, and 99.7% within 3 SD.

Limitations:

1. **Not robust:** SD is sensitive to outliers and non-normality.
2. **Scale-dependent:** SD is affected by the unit of measurement.
3. **Not suitable for skewed data:** Alternative measures like Interquartile Range (IQR) or Median Absolute Deviation (MAD) may be more suitable.

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Best practices:

1. **Use with mean:** SD is most informative when used with the mean.
2. **Check normality:** Ensure data is normally distributed before using SD.
3. **Use alternative measures:** Consider IQR or MAD for non-normal or skewed data.
4. **Contextualize:** Interpret SD in the context of the data and research question.

By understanding SD's strengths and limitations, you can effectively interpret variability in your data.

5. discuss different types of skewness with example and explain they affect data interpretation

15:06 ✓✓

Skewness refers to the asymmetry of a probability distribution. There are two main types of skewness:

1. Positive Skewness (Right-Skewed):

- Tail is longer on the right side
- $\text{Mean} > \text{Median} > \text{Mode}$
- Example: Income distribution in India, where most people earn average salaries (e.g., ₹20,000-₹50,000 per month), but a few individuals earn extremely high incomes (e.g., ₹1 crore+ per month).

2. Negative Skewness (Left-Skewed):

- Tail is longer on the left side
- $\text{Mean} < \text{Median} < \text{Mode}$
- Example: Distribution of exam scores in a class, where most students score high (e.g., 80-100%), but a few students score very low (e.g., 0-20%).

Effects on Data Interpretation:

1. **Mean:** Can be misleading in skewed distributions, as it's pulled towards the tail. For example, the average income in India might be skewed by billionaires like Mukesh Ambani.

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2. **Median:** More robust, provides a better representation of central tendency. For example, the median income in India might be a better representation of the average person's earnings.
3. **Mode:** Least affected, represents the most frequent value.

Impact on Analysis:

1. **Positive skewness:** May indicate presence of outliers or extreme values, which can impact analysis. For example, a few extremely high-income earners might skew the average income calculation.
2. **Negative skewness:** May indicate a ceiling effect or a lack of variability in the data. For example, if most students score high on an exam, it might indicate that the exam is too easy.

Understanding skewness helps:

1. Choose the right measure of central tendency.
2. Identify potential outliers or extreme values.
3. Select appropriate statistical tests and models.